ADHITHYAN S

Penetration Tester | CPENT v2 & CEH v13 Certified

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PROFESSIONAL SUMMARY

Penetration Tester with CPENT v2 and CEH v13 certifications, with hands-on, evidence-documented experience across Active Directory exploitation, binary/memory-corruption exploitation, firmware reverse engineering, web application testing, and Linux privilege escalation. Independently designed, built, and published an AI-powered phishing detection browser extension (WebShield.AI) to production on the Chrome Web Store and Microsoft Edge Add-ons Store, and developed WebRecon, an open-source 15-module web reconnaissance tool. Completed cybersecurity job simulations with Deloitte and TCS, and holds ISO/IEC 27001:2022 Information Security Associate certification. Seeking a penetration testing, red team, or web application security role.

TECHNICAL SKILLS

Penetration Testing Methodology: OWASP Top 10, SQL Injection, XSS, IDOR, SSRF, CSRF, RCE, Authentication Bypass, API Penetration Testing, JWT/OAuth2 Security, CVSS Risk Scoring, Evading Defense Mechanisms, Report Writing

Active Directory & Windows: LDAP enumeration, Kerberos AS-REP roasting, RODC assessment, Advanced Windows Attacks, credential/hash extraction (Impacket, NetExec), privilege escalation, credential cracking (John the Ripper, hashcat)

Binary Exploitation: Dynamic debugging and crash analysis (GDB), static disassembly/decompilation (Ghidra), exploit development and weaponization with pwntools, cyclic pattern offset analysis, exploit mitigation detection (checksec)

Web Application & CMS: Burp Suite, sqlmap, Gobuster, Joomscan, WPScan, Hydra

Firmware / IoT: Binwalk, SQLite3, file system and embedded credential extraction

Recon & Vulnerability Assessment: Subfinder, Amass, FFUF, Dirsearch, Nuclei, passive/active reconnaissance

Linux Privilege Escalation: SUID binary abuse, Linux capability abuse, sudo misconfiguration

Scripting & Automation: Python (security automation, attack automation, ML), Shell scripting, JavaScript

Cloud & Compliance: ISO/IEC 27001:2022 ISMS, IAM fundamentals, risk assessment

Platforms & Infrastructure: Kali Linux, Linux CLI, Pivoting and Double Pivoting (SSH SOCKS), Bypassing Filtered Networks, Manifest V3 extension development, REST API security, Nmap, Netcat

CERTIFICATIONS

- **Certified Penetration Testing Professional (CPENT) v2.** EC-Council. Issued Oct 2026, valid through Nov 2027.
- **Certified Ethical Hacker (CEH v13).** EC-Council
- **Advanced Diploma in Cyber Defense (ADCD).** Red Team Hacker Academy
- **ISO/IEC 27001:2022 Information Security Associate.** SkillFront
- **Cybersecurity Job Simulation.** Deloitte, via Forage
- **Cybersecurity Analyst Job Simulation.** TCS, via Forage. IAM and platform integration focus
- **CTF Achievement.** Technovally Hackfest 1.0

CPENT PRACTICAL EXAMINATION: TECHNICAL SUMMARY

- **Active Directory:** LDAP enumeration, AS-REP roasting (Impacket), hash cracking (John, hashcat), RODC credential exposure. 11/12 completed.
- **Binary Exploitation:** Crash analysis & exploit development with GDB, pwntools, Ghidra; built a working remote buffer-overflow exploit. 9/9 completed.
- **IoT / Firmware:** Reverse-engineered six firmware images (Binwalk, SQLite3), recovering embedded credentials and system details. 13/13 completed.
- **CTF-Style Linux Range:** SQL injection, file-upload RCE, CMS enumeration, privilege escalation (SUID, sudo, capabilities) across three hosts. 13/13 completed.
- **Web Application:** Access-control and logic-flaw exploitation. 2/6 completed.

KEY PROJECTS

WebShield.AI: AI-Powered Phishing Detection & Security Extension

2025 – Present

Founder & Sole Developer | Chrome Web Store & Microsoft Edge Add-ons Store

- Engineered and independently deployed a Manifest V3 browser extension for real-time phishing website detection, published and live on both the Chrome Web Store and Microsoft Edge Add-ons Store
- Trained a Random Forest ML model on URL structure and domain reputation signals to classify malicious URLs in real time, served via a Python backend + PHP API bridge
- Identified and engineered defenses against interstitial/popup-based attacks and tracker-based data exfiltration via query-parameter stripping
- Owned the full security-product lifecycle solo: research, model training, API integration, and deployment across two browser stores

WebRecon: Python Web Reconnaissance Tool

Open Source

github.com/4-dhi/-4-dhi-WebRecon

- Automated the full recon phase of a pentest engagement into one tool covering 15 tasks: footprinting & enumeration (WHOIS, DNS, SSL/TLS, robots.txt, sitemap discovery), application analysis (security headers, server fingerprinting), and infrastructure mapping (WAF detection, directory enumeration, proxy/SSRF checks)
- Built with Python 3, Requests, BeautifulSoup, Python-Whois, WAFW00F, and LXML; consolidates results into one report instead of 10+ separate tool outputs
- Validated end-to-end against OWASP Juice Shop, correctly flagging missing security headers, exposed configuration-style paths, and proxy infrastructure details

EDUCATION

Bachelor of Computer Application (BCA), University of Calicut | 2022 – 2025

Higher Secondary (Science), Memunda HSS | 2020 – 2022